C++ Cheat Sheet for Java Programmers

Your Name

Department / University

1. Basic Syntax

- Include headers: #include <iostream>, #include <cmath>, #include <string>
- Entry point: int main() { ... }
- Semicolons end all statements.
- Comments: // (single line), /* ... */ (multi-line)

2. Data Types

- Integers: int, long, long long
- Unsigned: unsigned int, unsigned long
- Floating point: float, double
- Boolean: bool
- Character: char
- String: std::string
- References: int& r = x; (alias to variable)

3. Variables and Constants

- Declaration: int age;
- Initialization: double pi = 3.14159;
- Constants: const int MAX_VALUE = 100;

4. Input and Output

- Input: cin » x;
- Output: cout « "Hello" « endl;
- Fixed precision: cout « fixed « setprecision(2) « value;

5. Operators

- Arithmetic: + - * / %
- Comparison: == != > < >= <=
- Logical: &&, ||, !
- Bitwise: &, |, ^, ~, «, »
- Assignment: += -= *= /= %=
- Increment / decrement: ++, –

6. Control Flow

- If/else: if(cond){...} else {...}
- Switch: switch(x){ case 1: ...; break; }
- Loops:
  - for(init; cond; inc){...}
  - while(cond){...}
  - do{...} while(cond);
- break, continue

7. Functions

- Syntax: return_type name(params){...}
- Overloading: same name, different parameters.
- Default parameters: int f(int x=0);

8. Arrays

- Declaration: int a[10];
- Access: a[0]

9. Pointers

- Declare: int* p;
- Address-of: &x
- Dereference: *p

10. Classes and Objects

- Class = data + methods
- Create objects: MyClass obj;

11. Strings

- Include: #include <string>
- Create: string s = "Hello";
- Methods: length(), substr(), find()

12. Standard Template Library (STL)

- Containers: vector, list, map, set
- Algorithms: sort(), find()
- Iterators for traversal

13. Memory Management

- Dynamic allocation: int* p = new int; to free memory delete p;
- Allocate an array: int *p=new int[20]; to free memory delete p[];
- Free memory: delete p;
- Smart pointers: unique_ptr, shared_ptr

Logical Expression Evaluation in C++

Logical Operators:

- AND: &&
- OR: ||
- NOT: !

Short circuit:

- false && expr right side not evaluated
- true || expr right side not evaluated

14. File Input/Output

- ifstream read files
- ofstream write files
- fstream both

15. Namespaces

- Avoid conflicts using namespaces.
- using namespace std;
- Or fully qualify: std::cout

16. Exception Handling

- try {...} catch(exception& e) {...}
- throw to signal errors

17. OOP Concepts

- Encapsulation, inheritance, polymorphism
- Access: public, private, protected
- Constructors / destructors

18. Templates

- Write generic code:

template <typename T> T max(T a, T b){...}

Key Differences from Java

- No garbage collection (manual memory management)
- Direct pointer manipulation
- Multiple inheritance allowed
- Operator overloading
- Lower-level control of hardware

C++ Hello World

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello␣World" << endl;
    return 0;
}
```